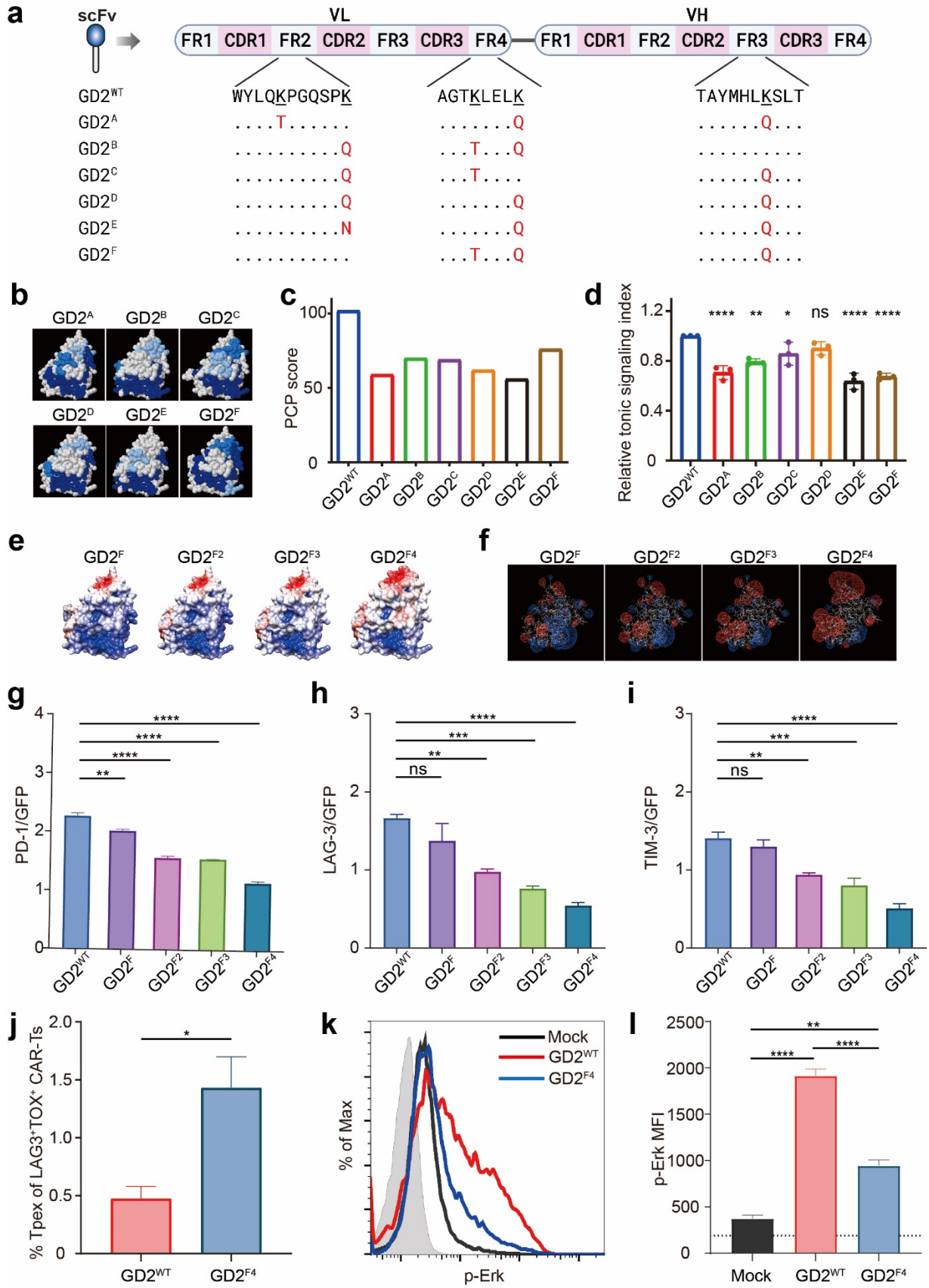


Figure S4



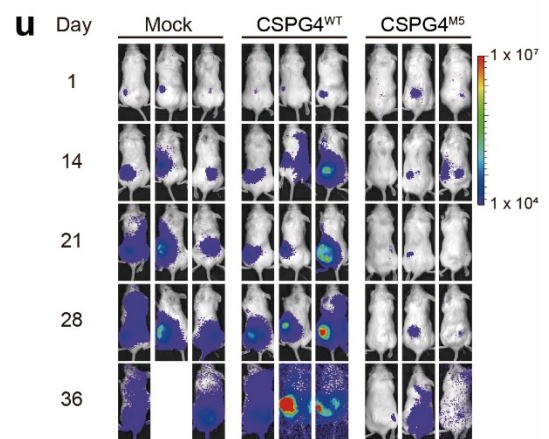
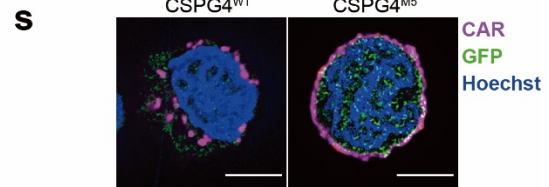
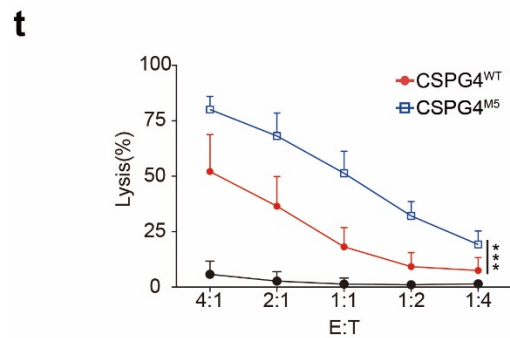
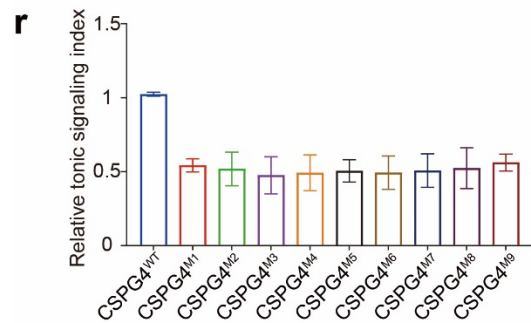
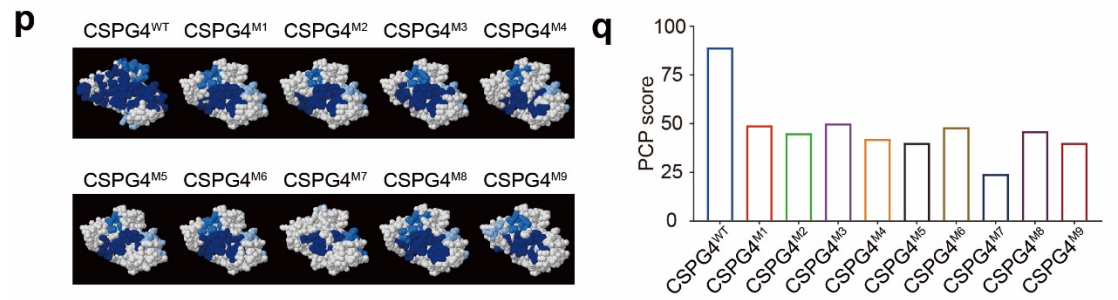
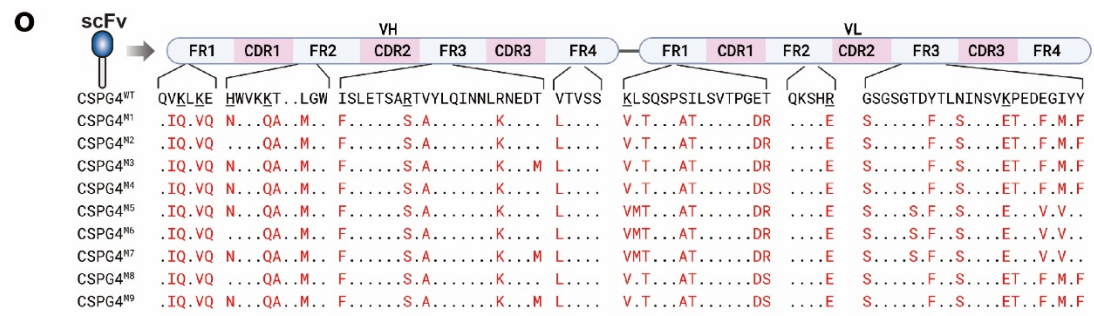
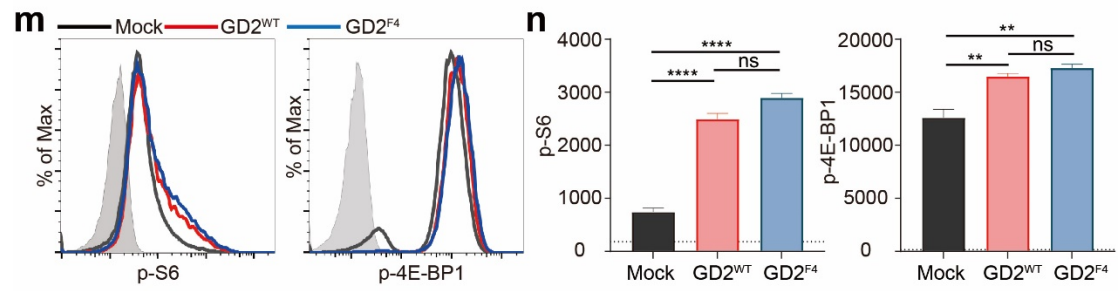


Figure S4. Tuning down PCPs on GD2 or CSPG4 CAR surface mitigates T cell exhaustion and improves CAR-T efficacy.

- (a) The sequences of initially optimized GD2.CAR variants.
- (b) The top three largest PCPs on modified GD2.CAR scFv surface displayed by the BindUP web server tool. Dark blue: the first-largest PCP; medium blue: the second-largest PCP; light blue: the third-largest PCP.
- (c) The calculation of PCP scores for the modified GD2.CAR scFvs.
- (d) The calculation of relative tonic signaling indexes for the modified GD2.CAR scFvs.
- (e) Electrostatics analysis of the optimized GD2.CAR scFv protein using APBS within UCSF Chimera. Blue, positively charged surface; red, negatively charged surface.
- (f) The electrostatic potential fields of the optimized GD2.CAR scFv observed in the Swiss-PDBViewer software.
- (g-i) Exhaustion markers including PD-1, LAG-3, and TIM-3 normalized by GFP expression in primary T cells expressing modified GD2.CAR were shown.
- (j) Percentage of TCF1⁺ progenitor exhausted cells in Lag3⁺TOX⁺ exhausted GD2^{WT} and GD2^{F4} CAR-T cells.
- (k-l) Phosphorylation of ERK in GD2^{WT} and GD2^{F4} CAR-T cells measured by FACS.
- (m-n) Phosphorylation of S6 and 4E-BP1 in GD2^{WT} and GD2^{F4} CAR-T cells assessed by FACS.
- (o) The sequences of optimized CSPG4.CAR variants.
- (p) The top three largest PCPs on optimized CSPG4.CAR scFv surface displayed by the BindUP web server tool. Dark blue: the first-largest PCP; medium blue: the second-largest PCP; light blue: the third-largest PCP.
- (q) The calculation of PCP scores for the optimized CSPG4.CAR scFvs.
- (r) The calculation of relative tonic signaling indexes for the optimized CSPG4.CAR scFvs.
- (s) Imaging analysis of CAR clustering on CSPG4^{WT} and CSPG4^{M5} CAR-T cells. Pink: CAR; green: CAR-IRES EGFP; blue: Hoechst. Scale bars, 5μm.
- (t) In vitro killing assay against CSPG4⁺ nasopharyngeal cancer cell line CNE-2 of CSPG4^{WT} and CSPG4^{M5} CAR-T cells.
- (u) Representative bioluminescence images of tumor burden after CSPG4^{WT} and CSPG4^{M5} CAR-T infusion over time.

Data are presented as means ± SEM; Comparisons were determined using unpaired student's t-tests (d, g-j, l, n) and two-way analysis of variance (t); * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$; ns not significant.